INCH-POUND
MIL-PRF-3098/133E
27 July 2005
SUPERSEDING
MIL-PRF-
3098/133D
27 August 1997

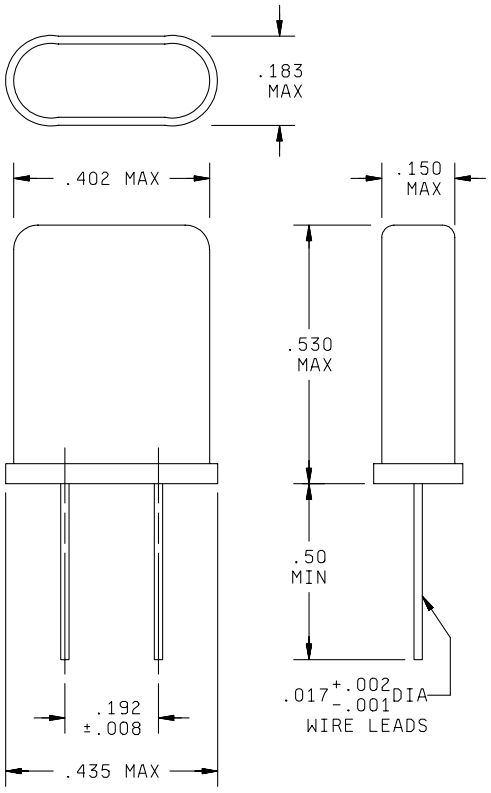
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR152/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 16 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.



Inches	mm
.001	.03
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.500	12.70
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR152/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 16.0 MHz to 61 MHz, inclusive

Frequency tolerance:

Primary operating temperature range: ±20 ppm.

Secondary operating temperature range: ±30 ppm.

Equivalent resistance: 40 ohms, maximum.

Mode of oscillator: Third mechanical overtone.

Operating temperature ranges (noncontrolled):

Primary: -40°C to +90°C, inclusive.

Secondary: -55°C to -40°C and +90°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 3.5 pF, maximum.

Unwanted modes: Method II, minimum unwanted-to-main mode resistance ratio of 3:1.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204, MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0811-10)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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